MIL-A-38140B(USAF)
1 August 1977
SUPERSEDING
MIL-A-38140A(USAF)
24 September 1963

MILITARY SPECIFICATION

ALTIMETER, PRESSURE AAU-16/A

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for an integrally lighted pressure altimeter, designated AAU-16/A.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-P-416	Plating, Cadmium (Electrodeposited)

MILITARY

MIL-P-116	Preservation-Packaging, Methods of
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-C-14806	Coating, Reflection Reducing, for Instrument Cover Glasses and Lighting Wedges
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENESS, Wright-Patterson AFB, OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6610

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MIL-G-81704 Glass, Aircraft Instrument, Lighting Wedge and Cover

STANDARDS

FEDERAL

FED-STD-595 Colors

MILITARY

MIL-STD-100 Engineering Drawing Practices
MIL-STD-129 Marking for Shipment and Storage
MIL-STD-130 Identification Marking of US Military Property
MIL-STD-143 Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-454 Standard General Requirements for Electronic Equipment
MIL-STD-470 Maintainability Program Requirements (For Systems and Equipment)
MIL-STD-471 Maintainability Verification/Demonstration/Evaluation
MIL-STD-781 Reliability Tests, Exponential Distribution
MIL-STD-794 Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810 Environmental Test Methods
MIL-STD-831 Test Reports, Preparation of
MIL-STD-859 Standard Calibration Table for Aeronautical Pressure Measuring Equipment
MIL-STD-889 Dissimilar Metals
MS33558 Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33638 Cases, Instrument, Flange Mounted, Aircraft
MS33649 Bosses, Fluid Connection, Integral Straight Thread

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The altimeter furnished under this specification shall be a product that has been tested and has passed the qualification tests herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

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3.3 Materials

3.3.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.3.2 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt fog, or atmospheric conditions likely to be met in storage or normal service.

3.3.3 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the altimeter except where magnetic materials are essential.

3.3.4 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.5 Corrosive fumes. Materials used in the interior construction of the altimeter shall not produce corrosive nor deleterious fumes under the test conditions specified herein.

3.3.6 Protective treatment. When materials are used that are subjected to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

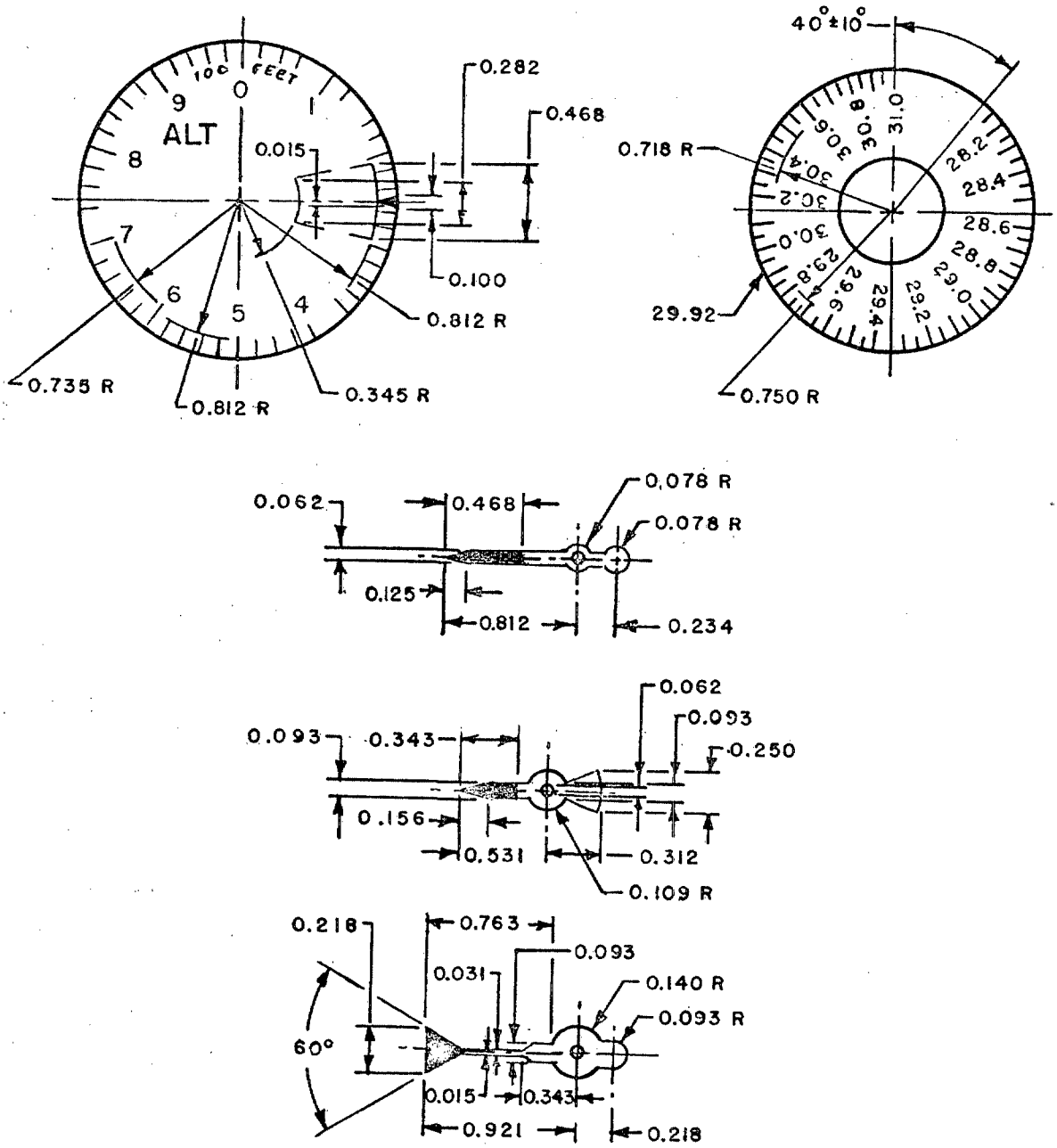
3.4 Design and construction

3.4.1 The altitude display shall be in accordance with figure 1 and shall provide a visual display from 0 to 100,000 feet altitude. The altimeter shall be designed to withstand pressures equivalent to 80,000 feet altitude without damage to the mechanism.

3.4.2 The altimeter shall be constructed to withstand the normal strains, jars, vibrations and other conditions incident to shipping, storage, installation and service.

3.4.3 Barometric setting system. A barometric setting system, consisting of a knob and sump in the altimeter, and located in the lower left-hand corner, shall be provided to permit the altimeter to be set to indicate zero altitude at any existing ground level pressure throughout a range of

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DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED,
TOLERANCES: DECIMALS ± 0.010

FIGURE 1. Dials and pointers.

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31 to 28.1 (inclusive) inches Hg. Suitable stops shall be provided in the barometric setting mechanism so that the scale cannot be read erroneously when the zero setting mechanism is moved beyond the range of 31 to 28.1 inches Hg. Rotation of the setting knob to either extreme shall not remove completely from view the numerals at each end of the scale.

3.4.3.1 Adjustment of setting. The zero setting system shall be provided with a suitable means to adjust or correct the relative setting of the altitude pointers and the zero setting indications, should this become necessary. This adjustment shall be accessible from the face of the altimeter when installed on the instrument board. The setting system shall incorporate a positive locking device to prevent inadvertent setting changes. Release of the locking device shall be accomplished by a slotted screw located on the altimeter bezel as shown on figure 2.

3.4.3.1.1 Zero set knob. Clockwise rotation of the zero set knob shall cause increasing readings of altitude as indicated by the pointers as well as increasing numbers indicated by the baro set subdial.

3.4.4 The altimeter shall be designed to facilitate determination of the instrument reading at the time of a crash provided this requirement does not significantly increase the item complexity or reliability (see 6.3.2).

3.5 Performance. The altimeter shall be capable of satisfactory operation when subjected to the following conditions:

a. Temperatures - Temperatures ranging from -62° to 71°C

b. Altitude - Pressures ranging from 31 inches Hg to 0.810 inch Hg (approximately 80,000 feet)

c. Vibration

(1) Vibration error not to exceed a pointer variation of 20 feet when subjected to an amplitude of 0.003 to 0.005 inch at frequencies from 500 to 3,000 cps

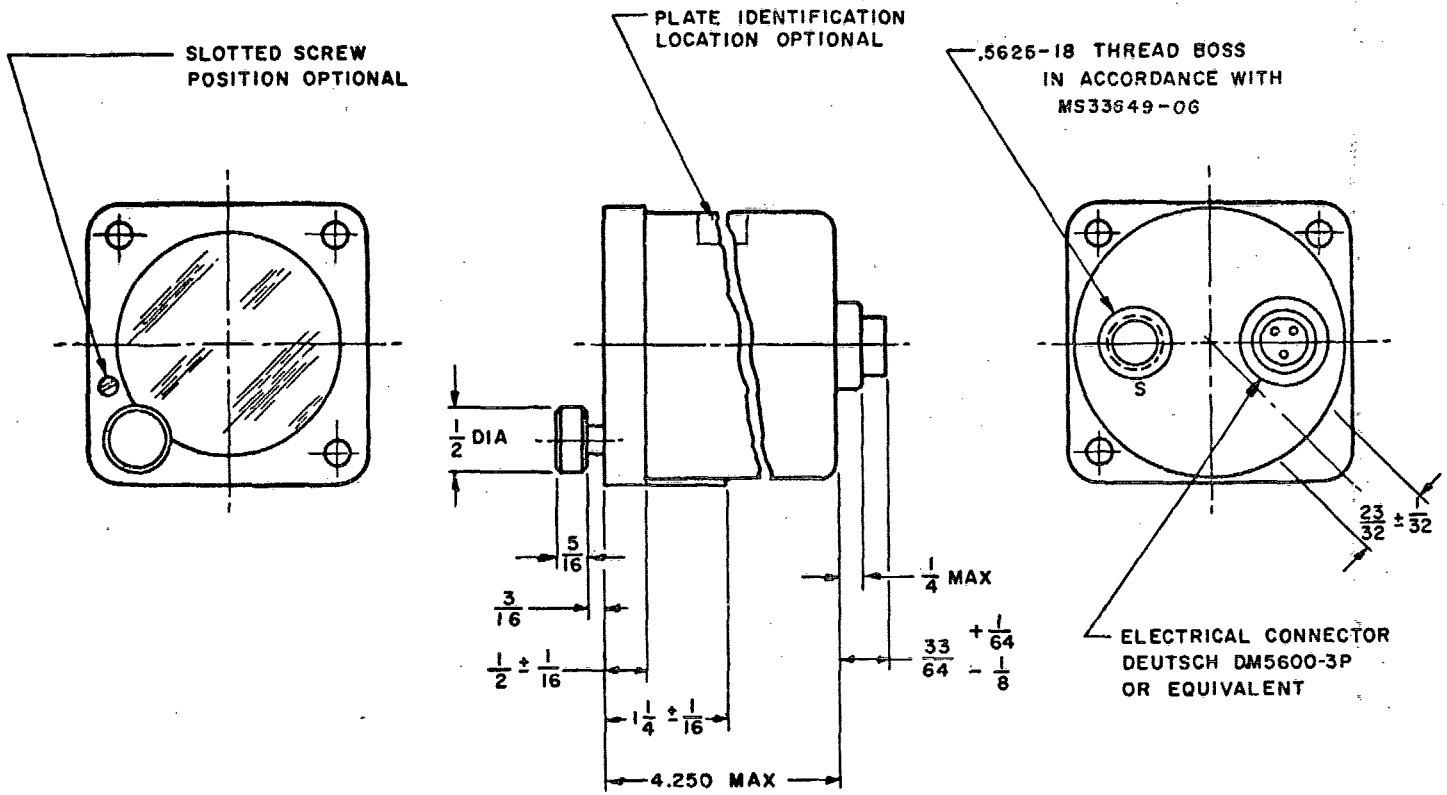
(2) Vibration failure within the specified limits when vibrated with a circular motion of 0.018 to 0.020 inch diameter at frequencies from 5 to 50 cps for a 3-hour period.

d. Acceleration - Forward acceleration forces of 2g

e. Humidity - Relative humidity up to 100 percent

f. Fungus - Fungus growth as encountered in tropical climates

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DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED

TOLERANCES: DECIMALS ±.005
FRACTIONS ± 1/64

FIGURE 2. Case.

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g. Salt fog - Exposure to simulated salt sea atmosphere

h. Dust (fine sand) - Exposure to sand and dust as encountered in desert areas

i. Case leakage - Case leakage not to exceed an indication of 100 feet

j. Simulated rainfall as encountered in any locale

k. Shelf life - 30 days storage.

3.5.1 Magnetic effect. When the indicator is held in various positions magnetically east or west of and not more than 12 inches from the center of a free magnet in a magnetic field having a horizontal intensity of 0.18 ±0.01 oersted, deflection of the magnet shall not exceed 1°.

3.5.2 Scale error. Scale error shall not exceed the tolerances specified in table I.

3.5.3 Position error. When the indicator is held in any position other than the normal operating position, the indicator reading shall not differ from the reading in the normal position by more than 20 feet.

3.5.4 Friction error. When the pressure is varied uniformly, the pointer shall operate smoothly, and the difference in readings before and after tapping shall not exceed the values specified in table II.

3.5.5 Reliability. The altimeter shall have a contract specified mean-time-between-failures (MTBF) of 1,200 hours. The reliability level shall be demonstrated in accordance with MIL-STD-781, test plan I.

3.5.6 Maintainability program. The contractor shall establish a maintainability program in accordance with MIL-STD-470. The following shall be considered in the design of the altimeter:

a. Minimization of complexity of maintenance tasks (calibration, adjustments, inspections, et cetera) by maximum use of simple design which induces optimum interchangeability and use of standardized equipment or commercial items

b. Optimum accessibility in all components requiring maintenance, inspections, removal, or replacement.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

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TABLE I. Scale errors.

Standard Altitudes	Equivalent Pressure mm Hg	Room Temperature Variation from Standard Altitude	-35° and 45°C Variation from Room Temp Reading	-54°C Variation from Room Temp Reading
0	760.00	±50	±50	±75
500	746.37	50		
1,000	732.93	70		
2,000	706.65	80		
3,000	681.14	90		
4,000	656.38	100		
6,000	609.05	110	125	150
8,000	564.52	120		
10,000	522.66	130		
12,000	483.35	150	175	270
14,000	446.47	175		
16,000	411.90	190		
18,000	379.53	210	275	420
20,000	349.25	225		
22,000	320.96	275		
25,000	282.03	300	375	560
30,000	225.69	325		
35,000	178.83	425	525	790
40,000	140.67	500		
45,000	110.62	625		
50,000	86.99	750	750	1,500
55,000	68.40	875		
60,000	53.79	1,000	1,000	2,250
70,000	33.29	2,500		

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TABLE II. Friction.

Altitude Feet	Tolerance ±Feet
1,000	70
2,000	70
3,000	70
4,000	80
10,000	90
16,000	100
20,000	115
25,000	130
30,000	150
35,000	175
40,000	200
50,000	250
60,000	450

3.7 Case. The altimeter case shall conform to MS33638 except as shown on figure 2.

3.7.1 Cover glass. The cover glass shall conform to MIL-G-81704. It shall be properly sealed and means shall be provided for replacement of the cover glass by removal of the bezel. The bezel shall be held in place by means of screws properly secured by lockwashers or similar devices of approved design. If the cover glass is used as part of the lighting system, a glass may be used that deviates from the glass specified herein to the extent required for the lighting function if it does not interfere with readability of the instrument.

3.7.2 Dial-to-cover-glass distance. The distance from the dial to the cover glass shall be the minimum possible and shall not exceed .25 inch.

3.7.3 Shipping plug. The static pressure connection shall be provided with a vented plug of acceptable material and design. The plug shall incorporate a vent consisting of a hole .0625 ±.0156 inch in diameter.

3.7.4 Case marking. The case shall be marked as follows in a location adjacent to the pressure connection:

"S" (for static connection).

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3.7.5 Anti-reflective coating. All visible glass surfaces shall be provided with a reflection reducing coating which shall meet the requirements of MIL-C-14806 and in addition shall withstand the environmental conditions specified herein.

3.8 Integral lighting. The altimeter shall be integrally lighted with both red and white circuits. The red lighting shall be in accordance with MIL-L-25467 and the white lighting shall be in accordance with MIL-L-27160. Lighting shall operate on 5V ac or dc. Lighting acceptability of production altimeters shall be based on visual comparison with a prime standard. The method of lighting shall be approved by the procuring activity prior to fabrication.

3.8.1 Prime standard. The prime standard lighting reference shall be prepared as follows: A minimum of 20 areas covering the entire face of the altimeter shall be measured for brightness level. A minimum of 10 areas covering the entire face of the altimeter shall be measured for color. The prime standard and its recorded overall brightness and color measurements shall be approved by the procuring activity. The standard shall be checked at least every 50 hours of operation to determine any change in brightness and color. Variations greater than ±20 percent in two or more areas shall require return of the standard for corrective action.

3.9 Dials

3.9.1 Numerals and lettering. The dials shall conform to figure 1. The style and proportions of the numerals and lettering shall conform to MS33558. Numerals shall distinctly indicate the graduation to which they apply.

3.9.2 Markings. The markings listed below shall be finished in lusterless white, Color No. 37875 of FED-STD-595.

Main Dial	Height or Length Inch ±0.010	Width of Line or Graduation Inch ±0.010
Dial numerals 0, 1, 4, 5, 6, 7, 8, 9	0.125	0.020
Minor graduation	0.125	0.015
Major graduation	0.187	0.031
Subdial pointer	See figure 1	
Lettering ALT	0.140	0.020
Notation 100 FT	0.078	0.010
Subdial		
Numerals	0.100	0.015
Major graduations	0.125	0.015
Minor graduations	0.093	0.010
29.92 mark	0.060	0.015

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3.9.2.1 The dial background and all markings not otherwise specified shall be finished in durable lusterless black, Color No. 37038 of FED-STD-595.

3.9.3 Visibility of dial markings. All graduations, numerals, pointers, and other markings on the dial shall be readable from any point within the frustrum of a cone, the side of which makes an angle of not less than 20° with the perpendicular to the dial and the small diameter of which is the aperture of the dial.

3.10 Pointers. The pointers shall be proportioned as shown on figure 1. The unshaded portion of the pointers shall be durable lusterless black, Color No. 37038 of FED-STD-595 and the shaded portion shall be lusterless white, Color No. 37875 of FED-STD-595.

3.11 Electrical connections. Electrical connections shall conform to figure 3. The electrical connector on the case shall be a Deutsch type DM5600-3P, or equal, and shall be located as shown on figure 2.

3.12 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.13 Weight. The weight of the completely assembled indicator, without connecting fittings, shall not exceed 1 pound.

3.14 Finishes and protective coatings

3.14.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

3.14.1.1 Dials, small holes, and case inserts need not be anodized.

3.14.1.2 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.14.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments shall be used in accordance with MIL-C-5541 in lieu of anodizing.

3.14.1.4 Castings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.14.1.5 When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.14.1.6 Parts in hermetically sealed containers need not be anodized.

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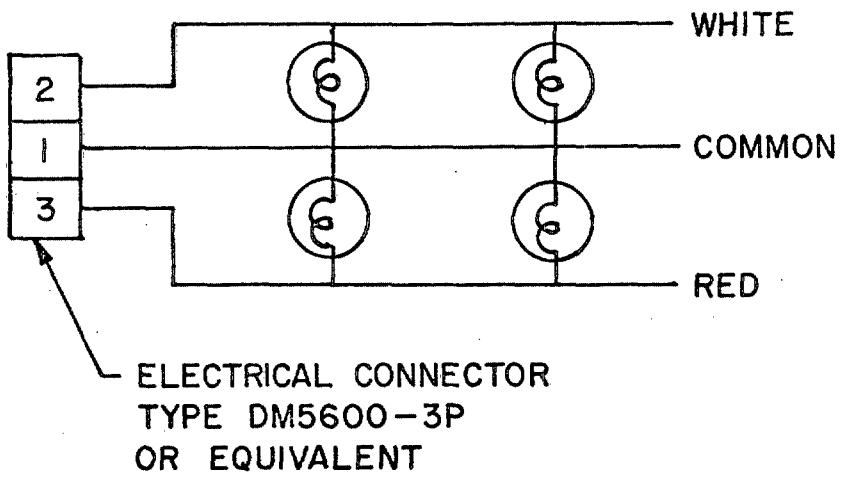


FIGURE 3. Wiring diagram.

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3.14.1.7 When the part is plated with tin over a copper flash, the part need not be anodized.

3.14.2 Cadmium plating. Cadmium plating shall be in accordance with QQ-P-416, type II or III, as applicable, and of a class that is adequate to achieve the degree of protection required.

3.15 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.16 Workmanship. The altimeter, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of marking of parts and assemblies and freedom of parts from burrs and sharp edges.

3.16.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the altimeter, they shall be held or limited accordingly.

3.16.2 Gears. Gear assemblies shall be properly aligned and meshed, and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustments, gear assemblies shall be free from backlash.

3.16.3 Cleaning. The altimeter shall be thoroughly cleaned and loose, spattered, or excess solder, metal chips, and other foreign material removed after final assembly. Burrs and sharp edges as well as resin flash that may crumble shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of altimeters shall be classified as follows:

a. Qualification tests

b. Quality conformance tests.

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4.3 Test conditions

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Vibration stand. Whenever a vibration stand is specified, it shall vibrate at any desired frequency between 300 and 3,000 cps and shall subject the altimeter to such vibration that a point on the instrument case will describe in a plane inclined 45° to the horizontal plane a circle not less than 0.003 or more than 0.005 inch diameter, or a circle of not less than 0.018 or more than 0.020 inch diameter, as specified in the tests.

4.3.3 Tapping. The altimeter shall be lightly vibrated or tapped before a test reading is taken. (See 6.3.1.)

4.3.4 Standard pressures. Standard pressures used in calibrating and testing the altimeter shall be as specified in MIL-STD-859.

4.3.5 Attitude. Unless otherwise specified, tests shall be made with the altimeter in normal operating position.

4.3.6 External connections. When external connections are required to simulate installed conditions, they shall consist of standard AN or MS connections and at least 10 feet of 1/4 inch copper tubing, coiled and so arranged to allow drainage of condensed vapors, and attached to the static pressure port of the altimeter.

4.4 Qualification testing

4.4.1 Qualification test samples. The qualification test samples shall consist of nine altimeters (see 4.4.3) representative of the production equipment. The samples shall be identified with the manufacturer's own part number and other information as required by the procuring activity.

4.4.2 Test report. The contractor shall prepare a test report in accordance with MIL-STD-831.

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4.4.3 Qualification. The qualification tests shall consist of all the tests specified in 4.6. The tests shall be conducted on two groups of samples in accordance with the following:

Group I	Group II
(Three Altimeters) All tests except realiability and shelf-life	(Six Altimeters) Reliability and shelf-life tests

4.5 Quality conformance tests. Quality conformance tests shall consist of:

a. Individual tests

b. Sampling tests.

4.5.1 Individual tests. Each altimeter shall be subjected to the following tests as described in 4.6:

a. Examination of product

b. Zero setting scale

c. Scale error at room temperature

d. Scale error at -35°C

e. Position error

f. Friction

g. Case leakage

h. Hysteresis

i. After effect

j. Scale error at 45°C

k. Individual tests of MIL-L-27160 and MIL-L-25467.

4.5.2 Sampling tests

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4.5.2.1 Sampling plan A. One altimeter shall be selected at random from each 100 or less produced on the contract or order and subjected to the following tests as described under 4.6:

a. Individual tests

b. Magnetic effect

c. Vibration error

d. Sampling plan A tests of MIL-L-27160 and MIL-L-25467.

4.5.2.2 Sampling plan B. Unless otherwise specified, three indicators shall be selected at random from the first 15 items on the contract and subjected to the following tests as described in 4.6:

a. Sampling plan A tests

b. Vibration failure

c. Low temperature exposure

d. High temperature exposure

e. Humidity

f. Fungus

g. Salt fog

h. Acceleration

i. Dust (fine sand)

j. Rain

k. Reliability and shelf life

l. Sampling plan B tests of MIL-L-27160 and MIL-L-25467.

4.5.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

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4.5.2.3.1 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure, but final acceptance of items on hand or later produced shall not be made until it is determined that items meet all requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. Prior to sealing, the indicator shall be inspected to determine compliance with the requirements of 3.4, 3.6, 3.8, 3.12, 3.13, 3.14, and 3.16. If these inspections are performed as part of the assembly inspections, they shall be documented during the equipment assembly and the documentation shall be furnished as test results for the test. After sealing, filling, and painting, the indicator shall be examined to determine compliance with the requirements specified herein with respect to materials, workmanship, markings, weight, dimensions, and visibility of the display components. The case shall be inspected for proper painting and workmanship and the connector shall be inspected to make sure that no pins are bent or damaged.

4.6.2 Zero setting scale. The altimeter shall be subjected to atmospheric pressure, tapped or vibrated lightly, and the pointer indication noted. After the series of readings has been recorded, the difference between the readings at the setting of 29.92 and all other settings shall be found, recorded, and compared with the values in the "Correct Difference" column specified in table III. The recorded values shall not differ from the correct differences by more than 25 feet at any setting.

TABLE III. Zero setting mechanism

Setting Pressure Scale	Correct Difference
28.1	1,727
28.5	1,340
29.0	863
29.5	392
29.92	0
30.5	531
20.90	893
30.99	974

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4.6.3 Scale error at room temperature. The zero setting system shall be set so the pressure scale will read 29.92. While the altimeter is subjected to atmospheric pressure, the reading on the pointers shall be taken and the error of the indication determined by comparison with the barometer. Without changing the zero setting, the altimeter shall be subjected successively to the pressures specified in table I. The reducing in pressure during the interval, as indicated by a standard mercurial barometer, between test points shall be made at a rate corresponding to an increase in altitude of approximately 3,000 fpm. The altimeter shall remain at the pressure corresponding to each test point at least 1 but not more than 5 minutes before a test reading is taken. The scale errors shall not exceed the tolerances specified in table I. With no vibration applied to the altimeter, the movement of the pointer shall be observed for any irregular motion. From 0 to 25,000 feet, the pointer shall not stop and jump more than 120 feet. From 25,000 to 40,000 feet, the pointer shall not stop and jump more than 450 feet. From 40,000 feet to maximum range, the pointer shall not stop and jump more than 850 feet.

4.6.4 Scale error at -35°C. The altimeter shall be maintained at -35° ±2°C for a 3-hour period. At the end of this period and while still at -35°C, the altimeter shall be subjected to the scale error at room temperature test at the test points specified in table I. The scale errors at these test points shall not exceed the tolerances specified. The friction requirements of 4.6.3 shall not apply to this test.

4.6.5 Position error. A reading shall be taken while the altimeter is in a vertical position (dial up) and after it has been vibrated. The altimeter shall then be rotated 90° to its normal operating position. Readings shall be taken after vibration in the 12, 3, 6, and 9 o'clock positions. None of the readings shall differ from the 12 o'clock reading by more than 20 feet. Vibration sufficient to remove all of the friction from the instrument shall be applied to the altimeter.

4.6.6 Friction. The altimeter shall be tested for friction at each of the test points listed in table II. The pressure shall be decreased to bring the pointer approximately to the desired reading, then held constant while two readings are taken; the first before the altimeter is tapped, the second after it it tapped. The difference in any two such readings shall be recorded as friction and shall not exceed the tolerances listed in table II. This test may be combined with the test specified in 4.6.3.

4.6.7 Case leakage. A vacuum shall be slowly applied to the static pressure port to produce an indication of approximately 40,000 feet. The vacuum line shall then be pinched off. After 1 minute, the indication shall not have decreased by more than 100 feet.

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4.6.8 Hysteresis. Not more than 15 minutes after the altimeter has been first subjected to the pressure corresponding to the upper limit of the scale in the test specified in 4.6.3, the pressure shall be increased at a rate corresponding to a decrease in altitude of approximately 3,000 fpm until the pressure corresponding to the first test point shown in table IV is reached. The altimeter shall remain at this pressure for at least 5 but not more than 15 minutes before the test reading is taken. After the reading has been taken, the pressure shall be further increased at the specified rate until the pressure corresponding to the second test point shown in table IV is reached. The altimeter shall remain at this pressure for at least 1 but not more than 10 minutes before the test reading is taken. After the reading has been taken, the pressure shall be further increased at the specified rate until atmospheric pressure is reached. The reading of the altimeter at either of the two test points shall not differ from the reading of the altimeter for the corresponding altitude in the test specified in 4.6.3 by more than the variation specified in table IV.

TABLE IV. Hysteresis.

Test Points	Altitude	Equivalent Atmospheric Pressure (mm Hg)	Permissible Variation
First point	25,000	282.03	150
Second point	20,000	349.25	150
Third point	------	ground level	60

4.6.9 After effect. Not less than 1 or more than 5 minutes after the completion of the test specified in 4.6.3, the pointers shall be returned to their original reading, corrected for any change in atmospheric pressure within the tolerance shown in table IV at the third test point.

4.6.10 Scale error at 45°C. The altimeter shall be maintained at 45° ±2°C for a 3-hour period. At the end of this period and while still at 45°C, the altimeter shall be subjected to the test specified in 4.6.3 at the test points specified in table I. The scale errors at these points shall not exceed the tolerances specified. The friction requirements shall not apply to this test.

4.6.11 Lighting. The indicator lighting circuits shall be properly connected and the indicator subjected to the individual tests of MIL-L-25467 and MIL-L-27160.

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4.6.11.1 Visual comparison. This test shall be conducted by qualified personnel only after these personnel have been tested for normal color vision and have spent at least 20 minutes in the dark to become dark-adapted. The prime standard and the production indicator shall be placed side by side at eye level in completely dark surroundings and viewed at a distance of approximately 2 feet in a sighting cone of 30° from normal to the surface of the cover glass.

4.6.11.1.1 Brightness. With 2.50 ±0.01V applied to the indicator under test and 2.40 ±0.01V and 2.60 ±0.01V applied consecutively to the lighting terminals of the prime standard, the pointer, numerals, markings, et cetera, of the indicator under test and the prime standard shall be compared for brightness. The two instruments shall also be compared for lighting uniformity. The brightness of the production unit shall be within the brightness settings (2.40 and 2.60V) of the prime standard. If the brightness is dissimilar in any individual area or the lighting of the indicator and prime standard is not uniform, the questionable areas of the production units shall be measured with a photometer in accordance with MIL-L-25467 and MIL-L-27160. If the production units fail this test, they shall be rejected and returned for rework. Preproduction units shall be tested by using the photometer and shall not be tested by use of the prime standard.

4.6.11.1.2 Color. The color comparison test shall be made as specified in 4.6.11.1.1 except 4.50 ±01V shall be applied to both the production unit under test and the prime standard. The two indicators shall be visibly identical in color.

4.6.12 Magnetic effect. The altimeter, not operating, shall be rotated in a vertical plane about a short-bar magnetic compass with the nearest part of the altimeter 5 inches from and magnetically east or west of the center of the compass. Starting directly under the compass, the altimeter shall be held in positions 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° from the initial position. At each of these positions, the altimeter shall be rotated on its own horizontal axis until it is in normal upright position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed 1°. This test shall be repeated with the altimeter at rated voltage.

4.6.13 Vibration error. The indicator shall be mounted on the vibration stand in its normal operating position and secured by flange screws positioned as on an aircraft panel. The indicator shall be subjected to vibration with a double amplitude from 0.003 to 0.005 inch at a frequency from 500 to 3,000 Hz, so that any point on the indicator case will describe the circular path of double amplitude specified, and inclined at an angle of 45° to the horizontal and dial planes. The natural frequency of the indicator shall not occur within this range. While the indicator is being

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vibrated, the maximum total spread of pointer vibration and the pointer variation from its original position shall not exceed 20 feet. The integral lighting shall be operating during this test and shall not flicker.

4.6.14 Sampling plan A lighting test. All sampling plan A tests of MIL-L-27160 and MIL-L-25467 shall be performed.

4.6.15 Vibration failure. The instrument shall be mounted in its normal operating position with a line joining the centers of the two upper mounting holes horizontal and the dial in the vertical plane. The static port shall be connected to a vacuum source and to a reference mercury barometer. An absolute pressure of 5 inches Hg (approximately equivalent to 40,000 feet) shall be applied to the static port. The instrument shall be vibrated with a circular motion of 0.018 to 0.020 inch diameter in a plane inclined 45° to the horizontal plane. The frequency of vibration shall be varied uniformly from 5 to 500 Hz and return once each hour for a 3-hour period. Following the vibration period, the altimeter shall meet the test specified in 4.6.3.

4.6.16 Low temperature exposure. The indicator shall be subjected to a low temperature test in accordance with MIL-STD-810, Method 502.1 for a period of 48 hours at -62° ±2°C. At the end of the 48-hour period, the temperature shall be raised to -54° ±2°C and shall be maintained at this temperature for 4 hours. While still at this temperature, the indicator shall be tested for scale error. After the indicator has returned to room temperature, it shall meet the individual tests.

4.6.17 High temperature exposure. The test specified in 4.6.16 shall be repeated except that the indicator shall be subjected to a high temperature test in accordance with MIL-STD-810, Method 501.1, procedure I. After the indicator has returned to room temperature, it shall meet the individual tests.

4.6.18 Humidity. The indicator shall be subjected to a humidity test in accordance with MIL-STD-810, Method 507.1, procedure I. The indicator shall then be subjected to and meet the individual tests.

4.6.19 Fungus. The indicator, including applicable external connectors, shall be subjected to a fungus test in accordance with MIL-STD-810, Method 508.1, procedure I. At the end of the test period, the indicator shall be examined to ascertain that no fungus growth has occurred. The indicator shall then be subjected to and meet the individual tests.

4.6.20 Salt-fog. The indicator shall be subjected to a salt-fog test in accordance with MIL-STD-810, Method 509.1, procedure I, for a period of 48 hours. At the end of the 48-hour period, there shall be no damage or deterioration to any part of the indicator. The indicator shall then be subjected to and meet the individual tests.

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4.6.21 Acceleration. The acceleration test shall be performed in accordance with MIL-STD-810, Method 513.2, procedure I. The forward acceleration shall be 2g as noted in table 513.2-3 of Method 513.2. After testing, the indicator shall be subjected to and meet the individual tests.

4.6.22 Dust (fine sand). The dust test shall be in accordance with MIL-STD-810, Method 510.1, procedure I. After testing, the indicator shall be subjected to and meet the individual tests.

4.6.23 Rain. The altimeter shall be subjected to a rain test in accordance with MIL-STD-810, Method 506.1, procedure II. Upon completion of the rain test, the unit shall be subjected to and meet the individual tests.

4.6.24 Reliability. A minimum of six altimeters shall be subjected to a reliability test in accordance with the following procedures: Each altimeter shall be subjected to a pressure cycling of a minimum of three cycles per hour for a minimum of 880 hours or 2,400 cycles consisting of a pressure variation equivalent to approximately 3/4 of full scale. Each pressure cycle shall consist of operation to maximum required altitude and return to ambient. The maximum scale value shall be maintained for not less than 2 minutes. Approximately 2 minutes at ambient pressure shall be allowed between each pressure cycle. During the fifth cycling period, the altimeter shall be vibrated at an amplitude between 0.009 and 0.011 inch and a frequency of 35 cps. The cycling period, number of hours or cycles, and cycling temperature shall be accomplished in accordance with table V. After each period of cycling, the altimeter shall be subjected to and meet the room temperature scale error test except that six test stations only, representative of the low range, midrange, and high range, shall be tested. Room temperature tolerances shall apply. During each scale error test, when any two test points exceed allowable tolerances by 20 feet or more, one malfunction shall be charged against this test. Upon completion of the final cycle, the altimeter shall be rested at ambient pressure for a period of 48 hours after which, if so designed, it may be reset by means of any exterior adjustment mechanism. The adjustment shall be limited to 100 feet. The altimeter shall then be subjected to and meet the individual tests. Failure to meet any one of the individual tests shall be considered a malfunction. For each five test points out of tolerance during the scale error test, one malfunction shall be charged. In case of malfunction of any kind, the malfunction shall be recorded and the test interrupted while repairs are made. The test shall then be continued from the point of interruption. If recalibration is required to pass the scale error test, an additional malfunction shall be charged against this test. The test shall demonstrate a contract specified MTBF of 1,200 hours in accordance with MIL-STD-781, test plan I.

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TABLE V.

Cycling Period	Hours or Cycles		Ambient Temp (°C)
1	150	450	25
2	100	300	55
3	150	450	25
4	100	300	-35
5	100	300	25
6	100	300	55
7	100	300	-35
8	100	300	25

4.6.25 Shelf life. After completion of the reliability test, the altimeters shall be placed in storage for a period of not less than 30 days. During the 30-day period, the instrument shall remain as dormant as practicable. The mechanism shall not be exercised other than that caused by normal ambient barometric pressure variations. The temperature shall remain essentially constant; however, normal room-temperature variations will be permissible. After the 30-day period, the instruments shall again be subjected to and meet the individual tests. In addition to meeting the tolerances specified for scale error, the readings shall not vary from the original scale error readings by more than the values specified in table VI.

4.6.26 Sampling plan B lighting tests. All sampling plan B tests of MIL-L-27160 and MIL-L-25467 shall be performed.

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TABLE VI. Scale error variation.

Standard Altitude (Feet)	Scale Error - Variation Tolerances (Feet)
0	30
500	30
1,000	30
1,500	35
2,000	35
2,500	35
3,000	35
4,000	40
5,000	45
6,000	45
8,000	50
10,000	55
12,000	60
14,000	65
15,000	70
16,000	70
18,000	75
20,000	80
22,000	85
25,000	90
30,000	105
35,000	120
40,000	130
45,000	140
50,000	155
55,000	180
60,000	405
70,000	655
80,000	900

5. PACKAGING

5.1 Preservation-packaging. The altimeter shall be packaged in accordance with MIL-STD-794, level A or C as specified in the contract. For level A packaging, the method of preservation shall be in accordance with MIL-P-116, method 1A-5 without preservation compound, using reusable metal containers.

5.2 Packing. The altimeter shall be packed in accordance with MIL-STD-794, level A, B, or C, as specified in the contract.

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5.3 Marking. The interior and exterior containers shall be marked as specified in MIL-STD-794 with the following precautionary markings appearing conspicuously on two opposite sides of each interior package and each shipping container:

FRAGILE - DELICATE INSTRUMENTS HANDLE WITH CARE.

6. NOTES

6.1 Intended use. The AAU-16/A altimeter covered by this specification is intended for use as standby indicator in high performance aircraft to give accurate indications of altitude.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When sampling plan B tests are not to be conducted (see 4.5.2.2)

c. Levels of packaging and packing required.

6.3 Definitions

6.3.1 Tapping. The phrase TAPPED LIGHTLY is defined as the application of a light vibration sufficient to overcome that amount of residual friction normally associated with good quality aircraft instruments.

6.3.2 The requirements specified in 3.4.4 are intended to facilitate aircraft crash investigation through minor additions to standard indicator and transmitter techniques. Complex mechanisms to meet the requirement are not desired; rather, techniques such as fluoroscent-luminescent paint on the back side of pointers, index marks on internal moving parts, impalement spikes, or like accessories are preferred.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this

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specification. The activity responsible for the Qualified Products List is ASD/ENAID, Wright-Patterson AFB, OH 45433 and information pertaining to qualification of products may be obtained from that activity.

Custodian:
Air Force - 11

Review activity:
Air Force - 99

Preparing activity:
Air Force - 11

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